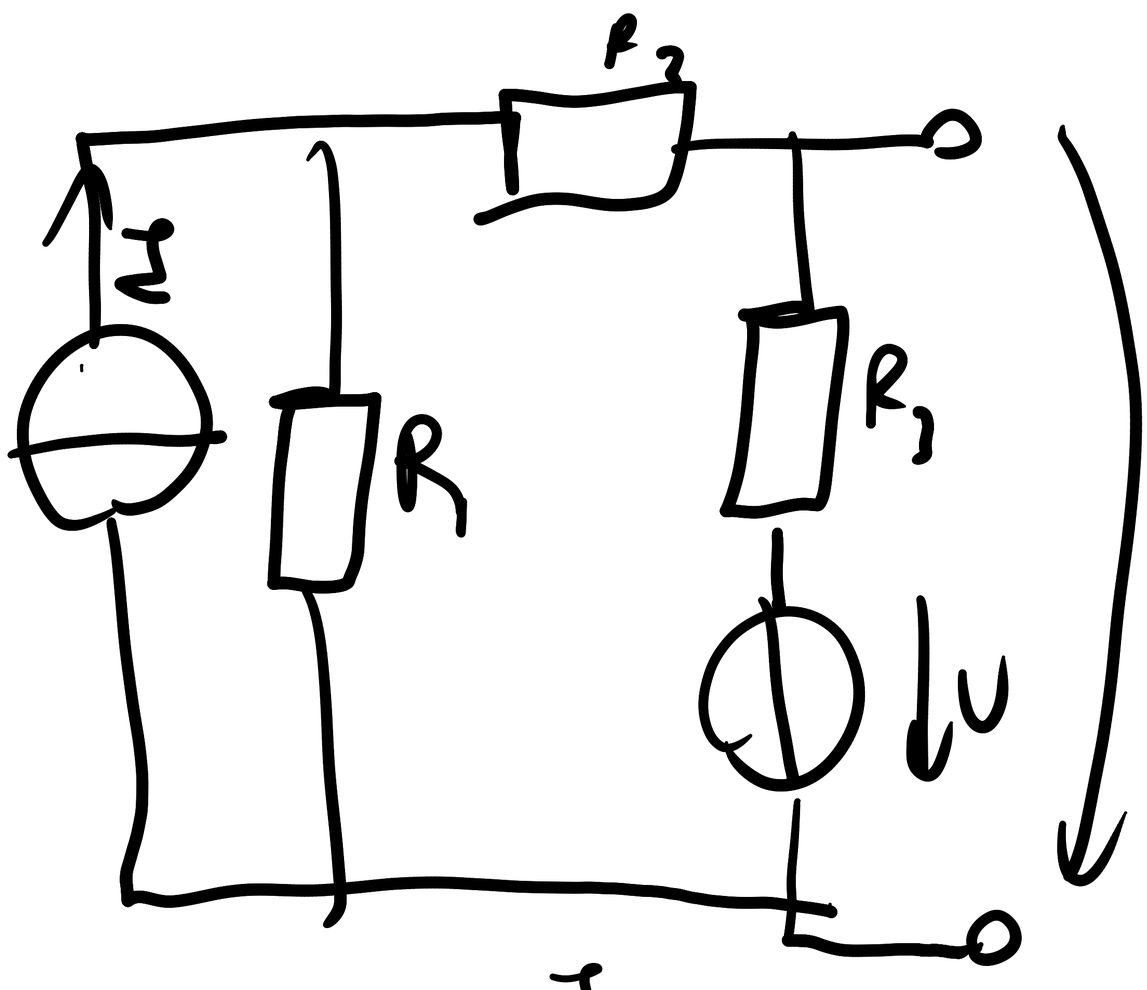


$$R_{23} = R_2 + R_3$$

$$\Rightarrow U_0 = R_3 \cdot I_2$$

$$I_2 + I_1 = I$$

$$U_0 = \boxed{I_1 R_1 = I_2 R_2 + I_2 R_3}$$

$$I_1 (R_1) - I_2 \cdot (R_2 + R_3) = 0$$

$$\begin{bmatrix} 1 & 1 \\ R_1 & -R_2 - R_3 \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \end{bmatrix} = \begin{bmatrix} I \\ 0 \end{bmatrix}$$

R_1 R

$$I_1 R_1 - I_2 (R_2 + R_3) = 0$$

$$I_1 + I_2 = I$$

$$I_1 = I - I_2$$

$$(I - I_2) R_1 - I_2 (R_2 + R_3) = 0$$

$$R_1 I - I_2 (R_1 + R_2 + R_3) = 0$$

$$R_1 I = I_2 (R_1 + R_2 + R_3)$$

$$\bar{I}_2 = I \cdot \frac{R_1}{R_1 + R_2 + R_3}$$

$$I_1 \cdot x_{11} + I_2 \cdot x_{12} = c_1$$

$$I_1 \cdot x_{12} + I_2 \cdot x_{22} = c_2$$

$$A_1 I_1 = c_1$$

$$A_2 I_2 = c_2$$

$$A_1 x_1 = b_1$$

$$A_2 x_2 = b_2$$

$$x_1 + x_2$$

$$(A_1 + A_2)\,x = (b_1 + b_2)$$

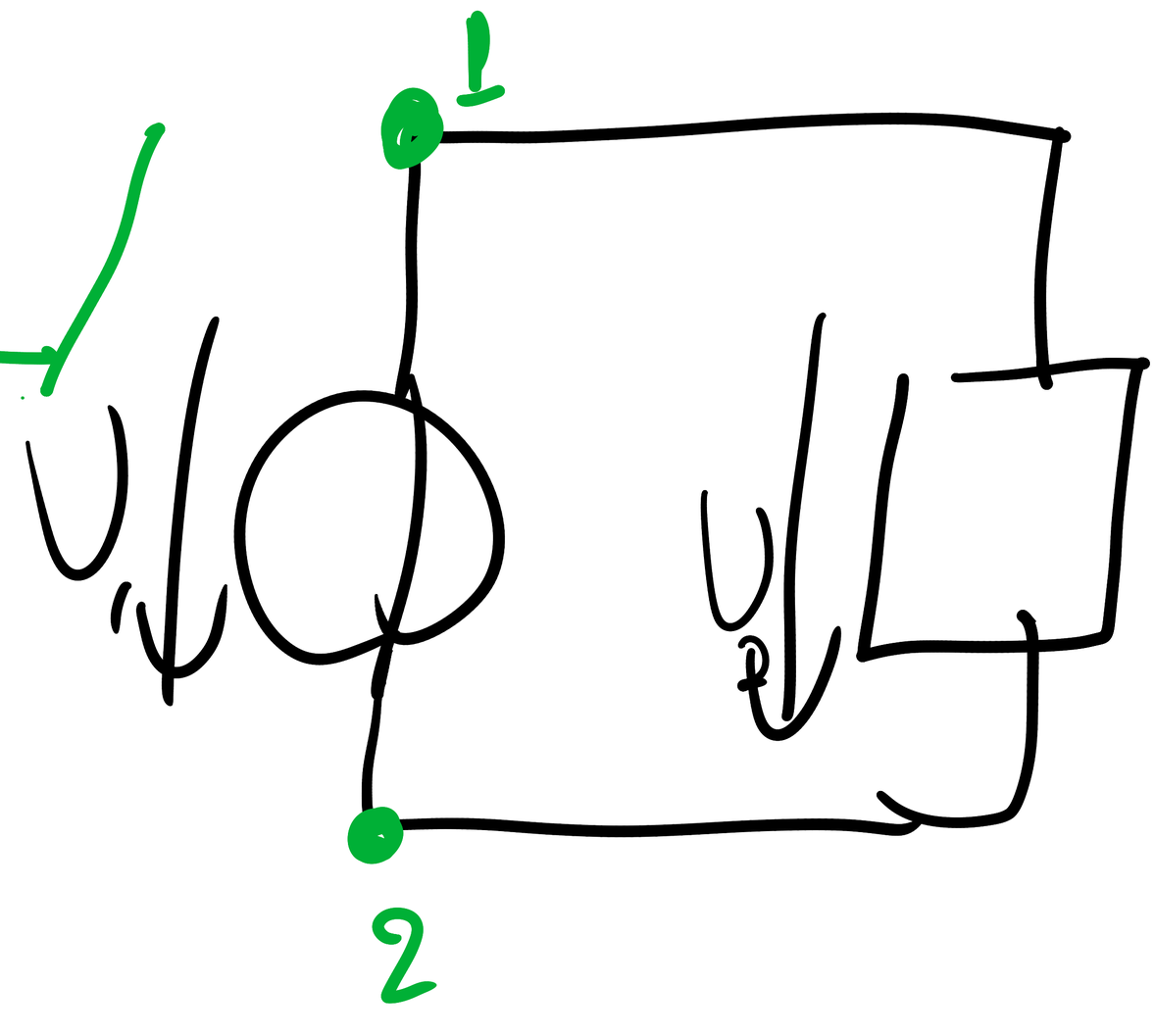
$$U_0 = U_1$$

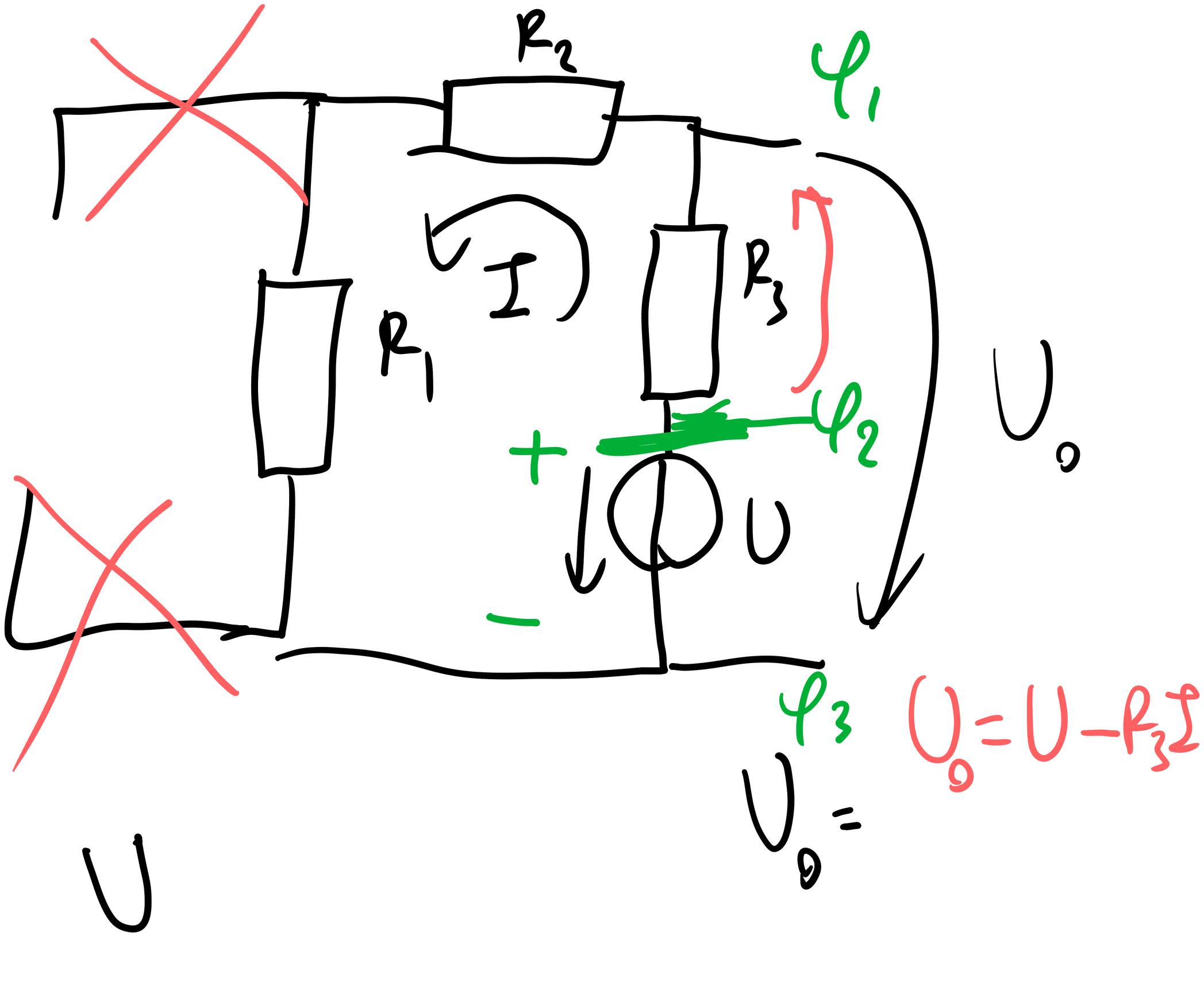
$$\frac{1}{\tau} \oint_{-}^{+} U$$

$$\varphi_\infty = 0$$

$$\varphi_x = \int_x^\infty \vec{E} \cdot d\vec{s}$$

$$\varphi_1 - \varphi_2 = U_1 = U_2$$





$$I(R_1+R_2+R_3)=U$$

$$U_0=\varphi_1-\varphi_3$$

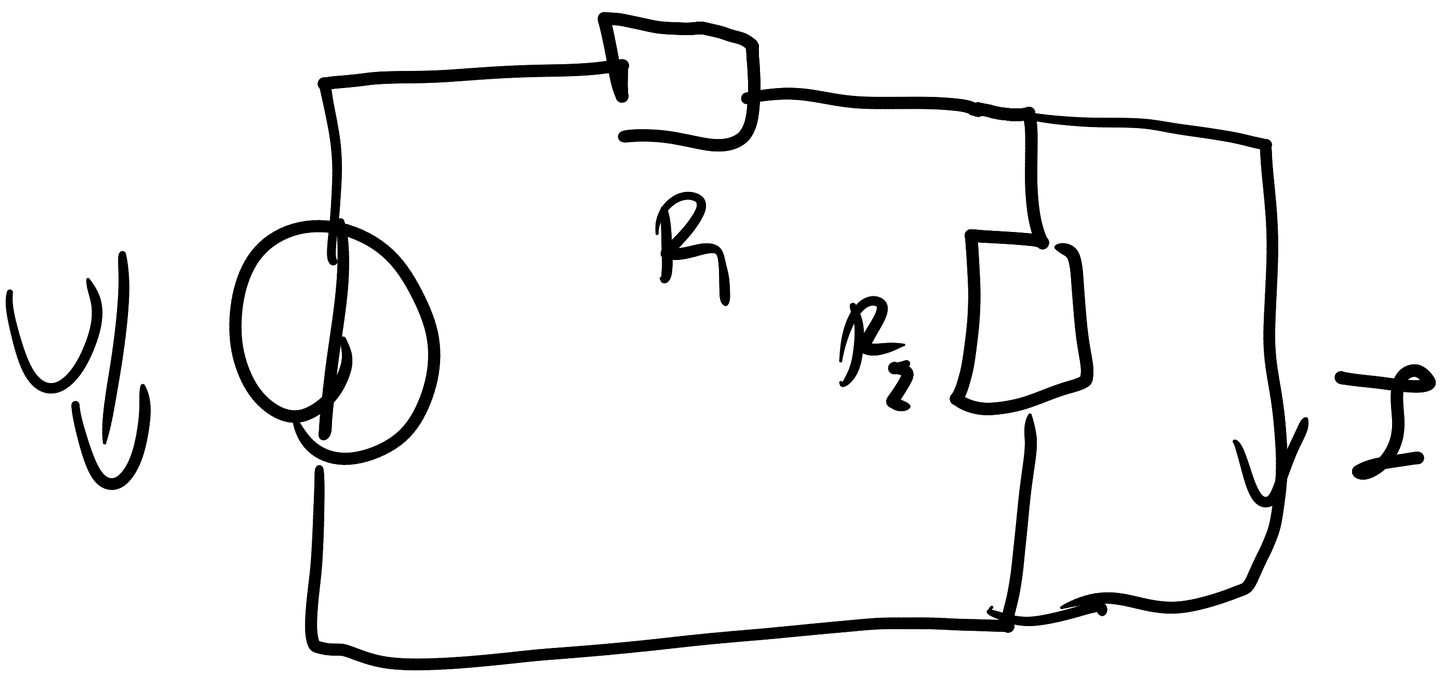
$$V_{R_3}=\varphi_1-\varphi_2$$

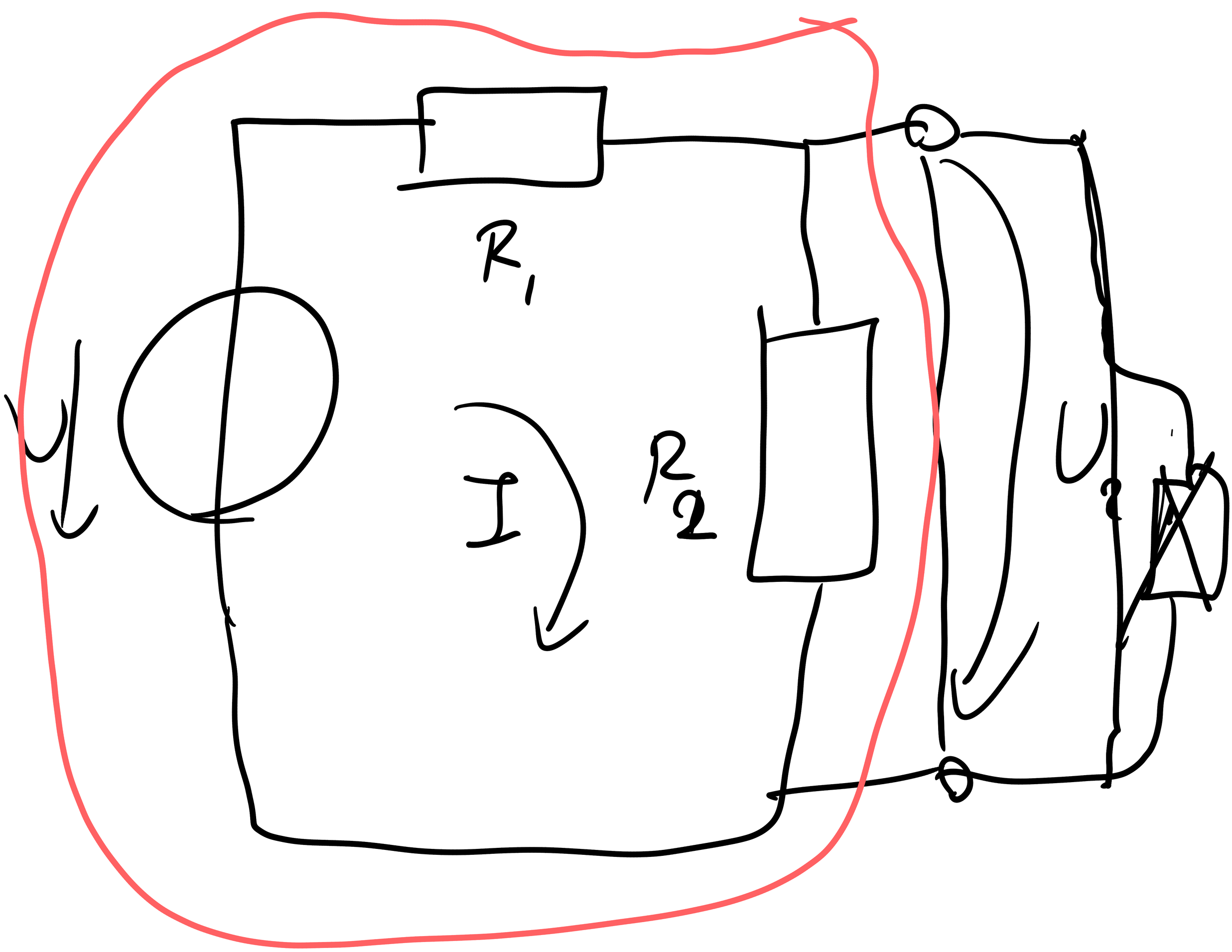
$$=-IR_3$$

$$I=\frac{U}{R_1+R_2+R_3}$$

$$U_0 = U - \frac{U R_3}{R_1 + R_2 + R_3}$$

$$= U\left(\frac{R_1 + R_2 + R_3}{R_1 + R_2 + R_3} - \frac{R_3}{R_1 + R_2 + R_3}\right)$$



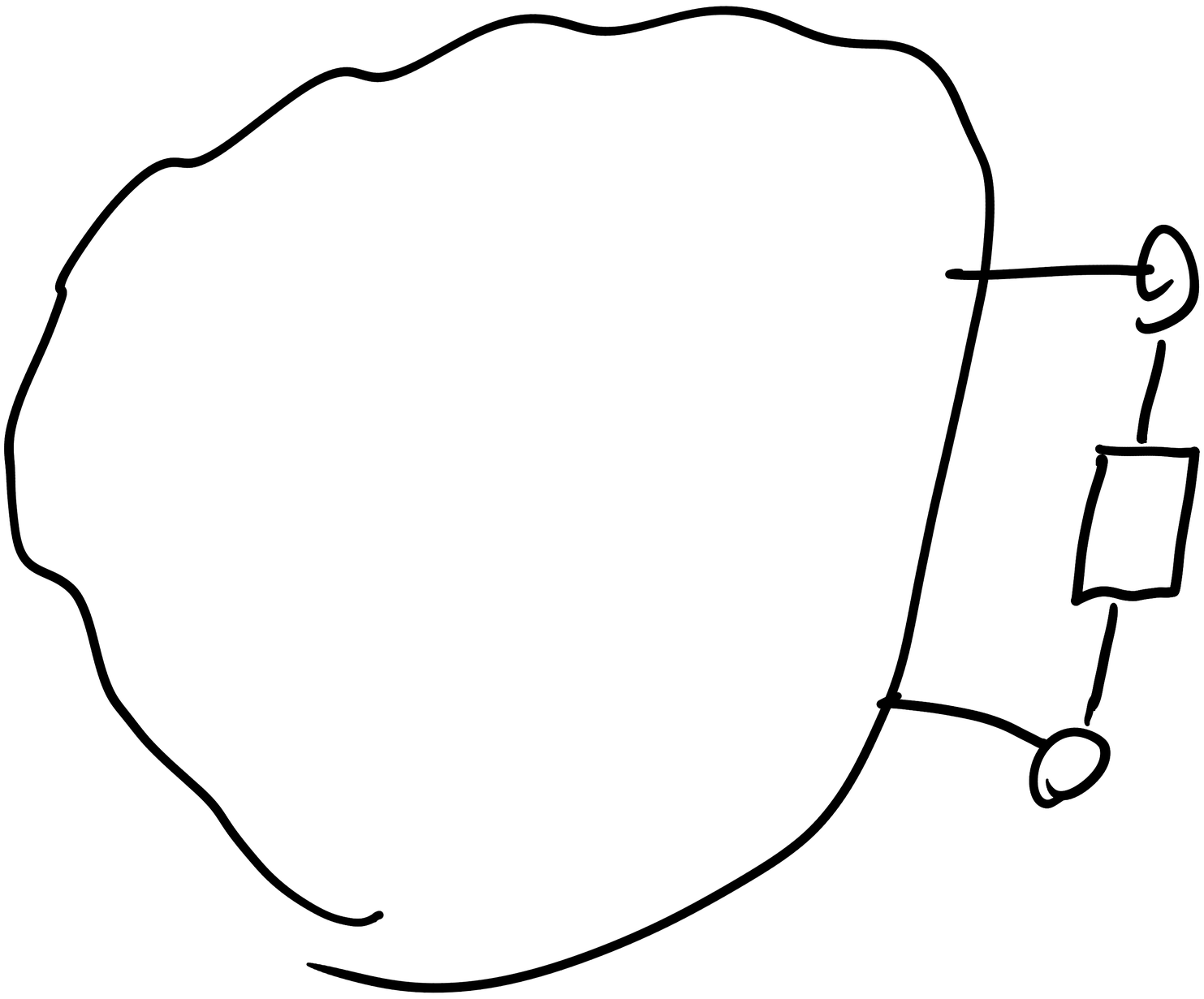


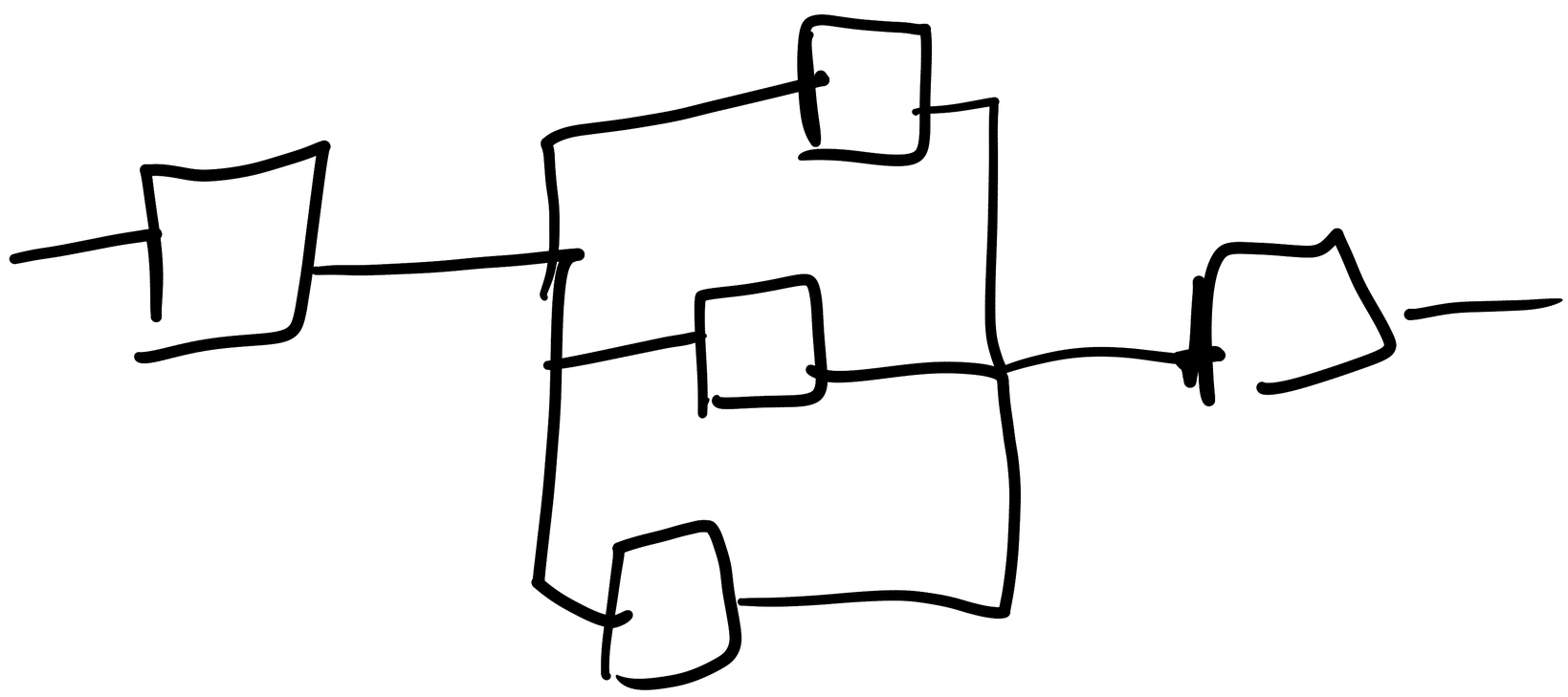
$$U_2 = U - IR_1 = IR_2$$

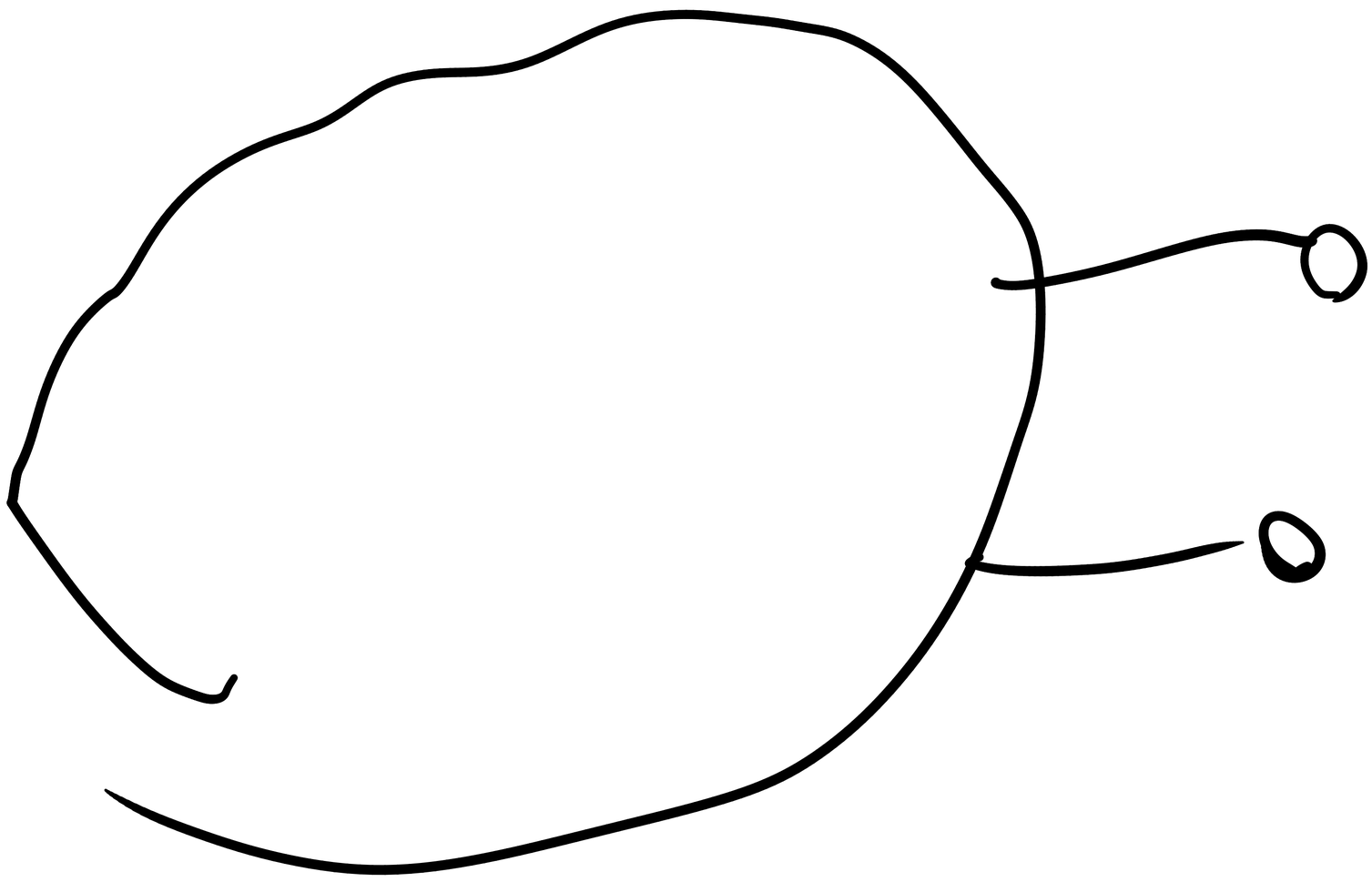
$$I(R_1+R_2)U$$

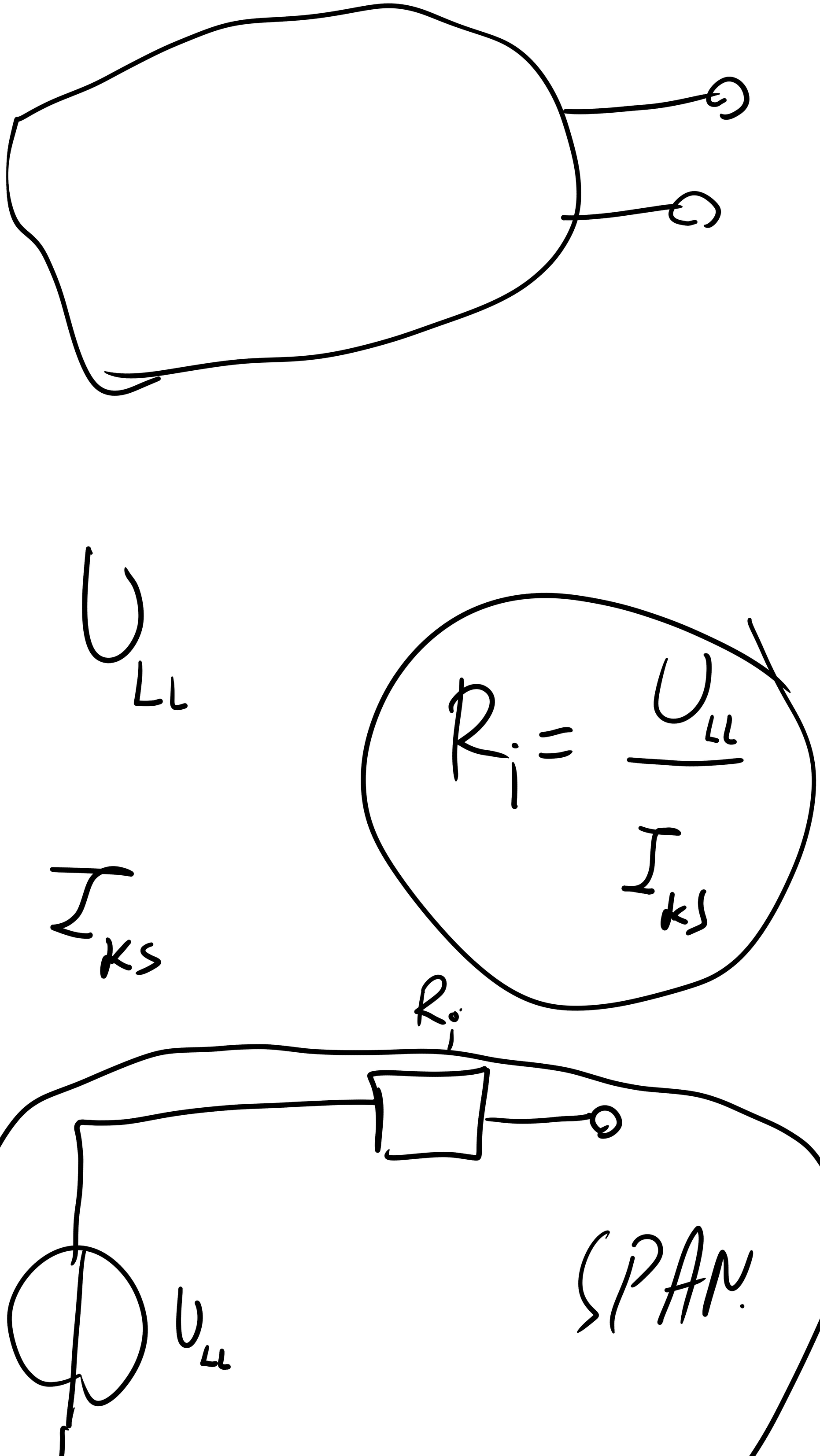
$$I = \frac{U}{R_1+R_2}$$

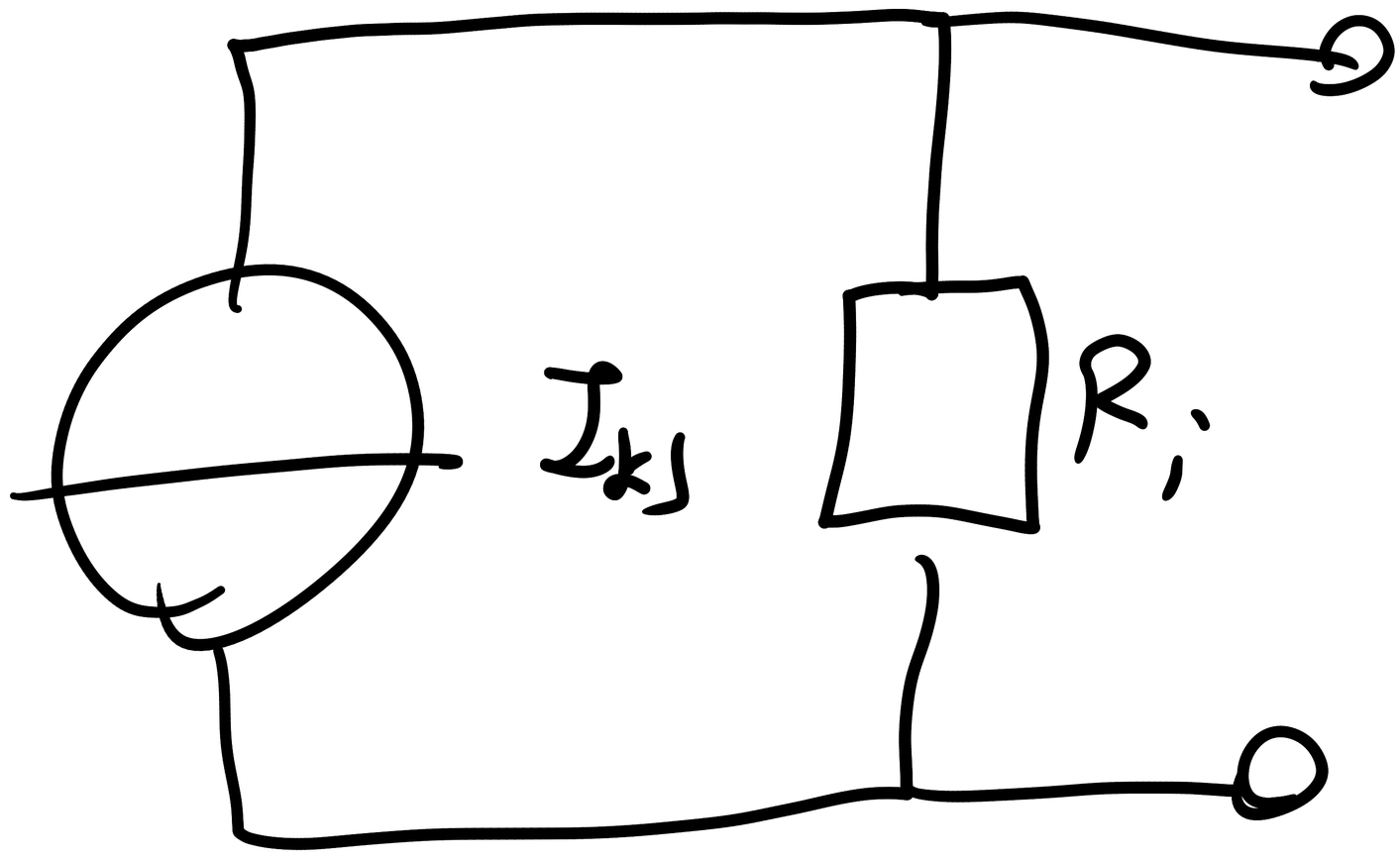
$$U_2 = \frac{R_2}{R_1+R_2} \cdot U$$

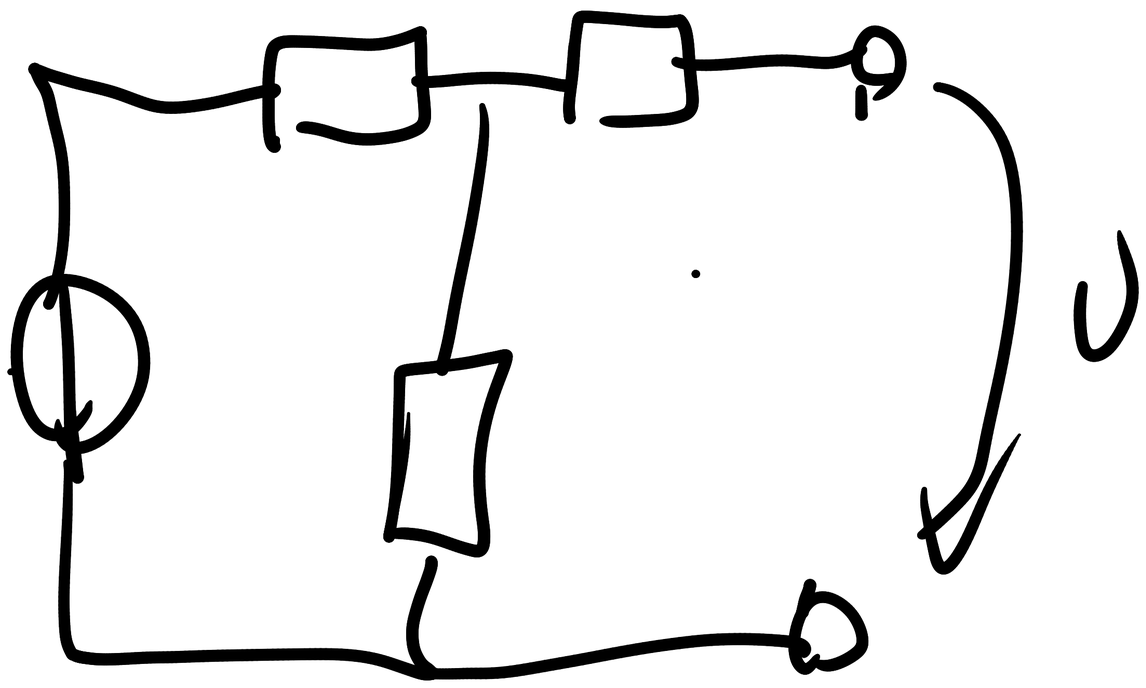


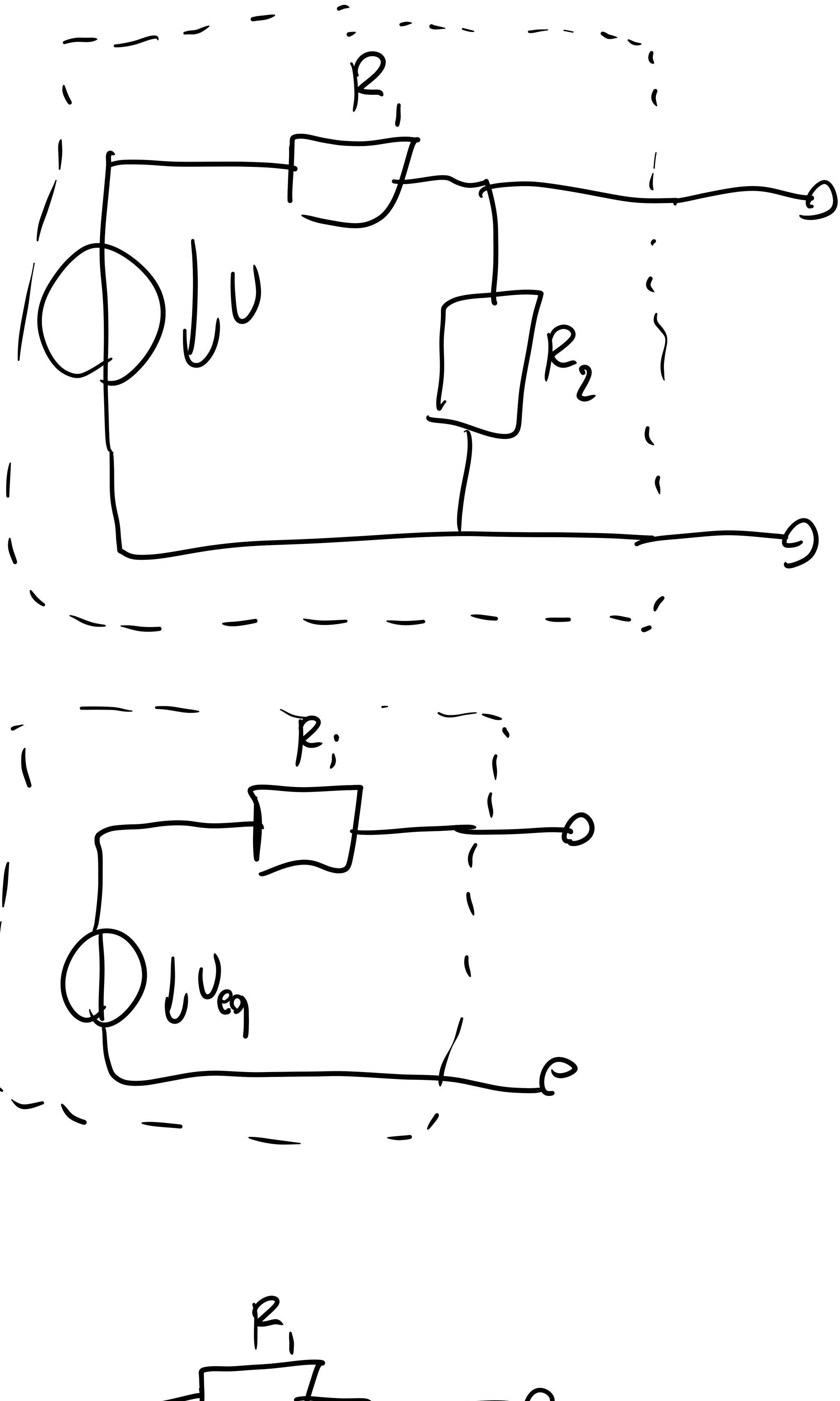


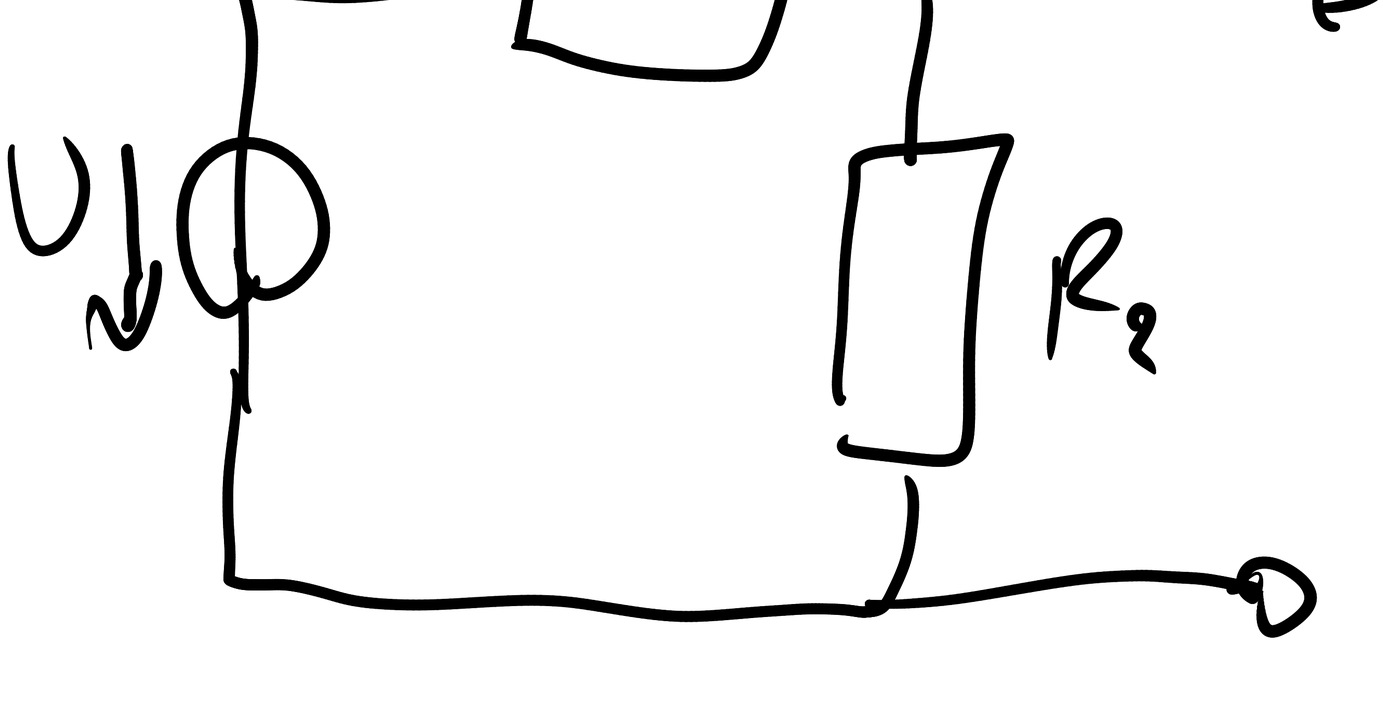



U_{LL}
$R_i = \frac{U_{LL}}{I_{KJ}}$
I_{KS}
R_i
U_{LL}
SPAN


I_{kj}
R_j

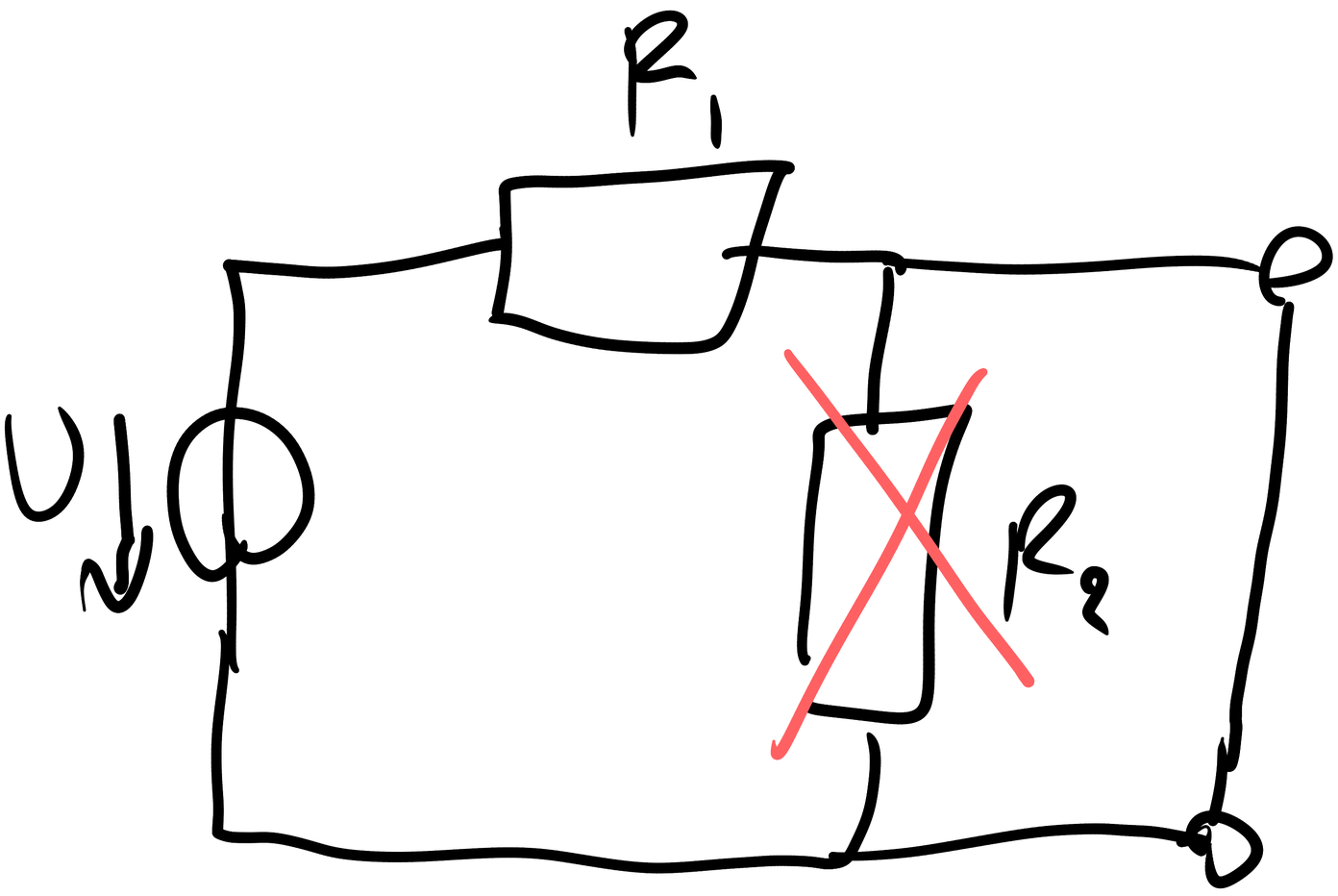



R1
U
R2
Ri
Ueq
R1



$$U_{L1} = \frac{R_2}{R_1+R_2} U$$

$$\eta = \frac{P_{oute}}{P_{total}}$$

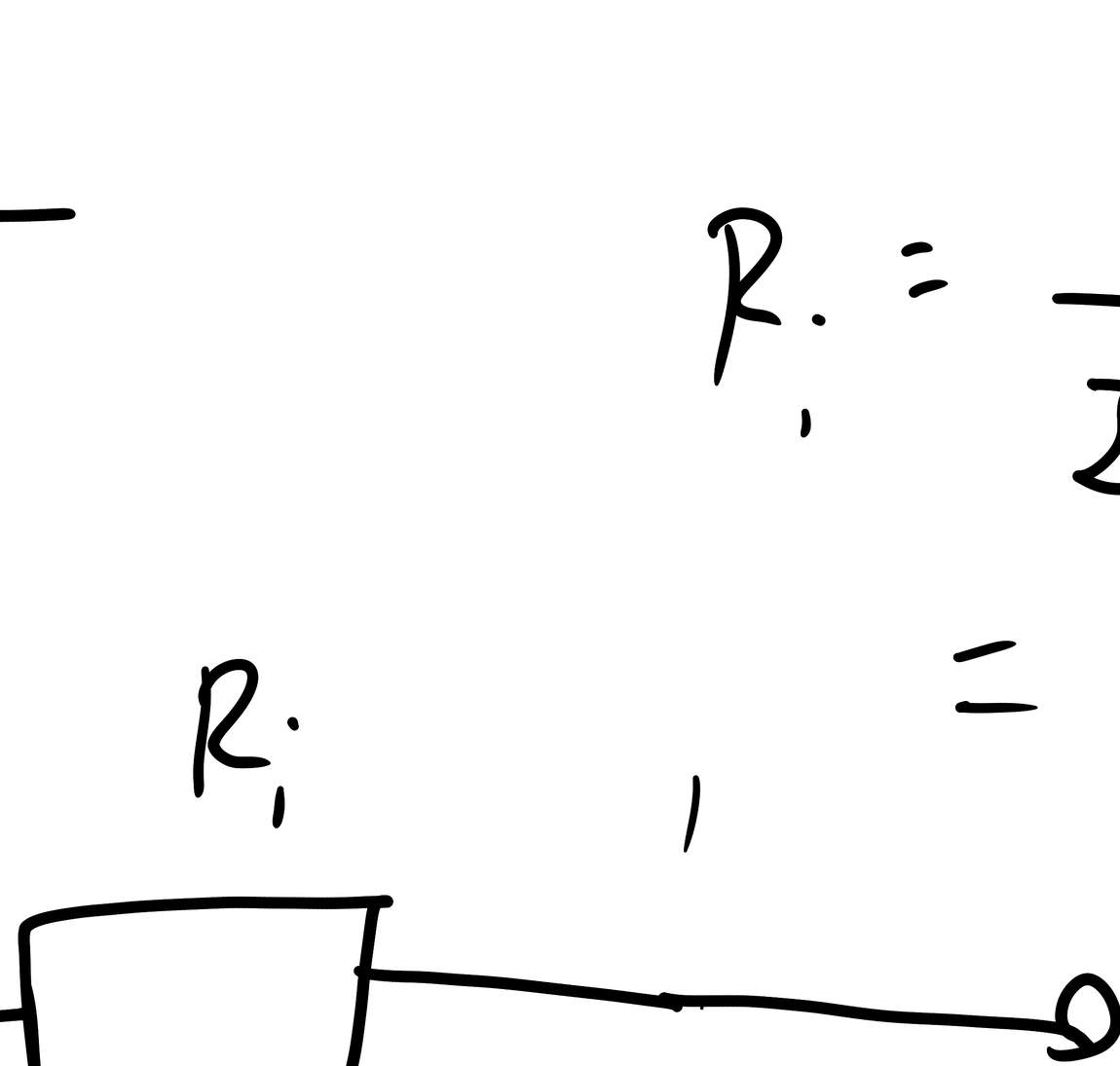


$\eta \to 1$

$\to [0, 1]$

$$I_{KS} = \frac{U}{R_1}$$

$$R_i = \frac{U}{I} = \frac{R_2 R_1}{R_2+R_1}$$



$U_{eq} = \frac{R_2}{R_1+R_2} U$

$ax+b$

